UGTS-KC 3.9.3

K-KIJ-T GROVE IQ FIELD EDITION

Formal Distance-Field, Procedural Signal and Bounded Ray-Query Expansion

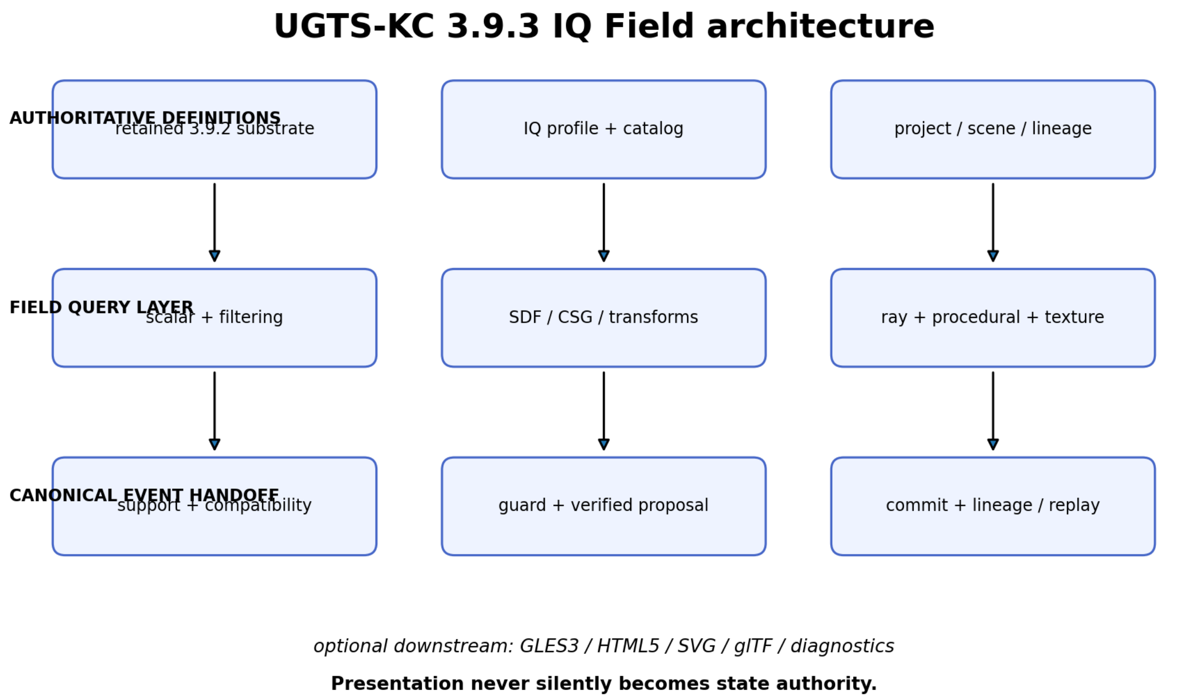

Figure 1. The IQ Field layer is additive and returns every state-changing result to the retained UGTS event authority.

Release field	Value
Prepared for	Tom Klootwijk
Requester-supplied identifier	NL200678942
Requester-supplied date of birth	10-07-1990
Version / date	3.9.3 / 28 August 2026
Release schema	ugts-kc-iq-field-profile-3.9.3
Mechanism range	M001-M609 by source/catalog

ATTRIBUTION AND EVIDENCE BOUNDARY

The name, identifier, date of birth and signature-edition label were supplied by the requester and are not independently verified. This package is a technical design and executable reference artifact. It is not legal proof of identity, authorship, ownership, patentability, priority, scientific validation or physical-GPU performance.

Contents and Executive Summary

Executive summary

Version 3.9.3 retains the complete supplied 3.9.2 K-Kij-T Grove package and adds a finite IQ Field profile distilled from the publicly indexed Inigo Quilez article inventory. The new layer covers typed remapping functions, exact and bounded distance fields, CSG, domain identity, analytic ray intersections, bounded sphere tracing, deterministic procedural signals, analytic filtering, projection and color helpers, plus Android/GLES integration. It does not replace the retained scene graph, geometry compiler, vector runtime, mobile 3D project schema, deterministic commit path, replay or Grove application.

RELEASE RESULT

A single versioned IQ field profile can be validated in JSON, evaluated by dependency-free Python, compiled as C++20 host/native helpers, embedded in GLSL ES/WGSL sources, exercised by a deterministic raymarched demonstration and exported with the existing Android project generator. Every iterative path has explicit budgets and terminal status.

Measured expansion

Metric	3.9.2 baseline	3.9.3 result
Combined mechanism range	M001-M529	M001-M609
Engineering catalog	M198-M529	M198-M609 (412 rows)
Automated tests	276	358 (82 new)
Article/update survey	not present	135 entries
IQ mechanisms	not present	80 (M530-M609)
Native field helper	not present	Python + C++20 + GLSL ES + WGSL

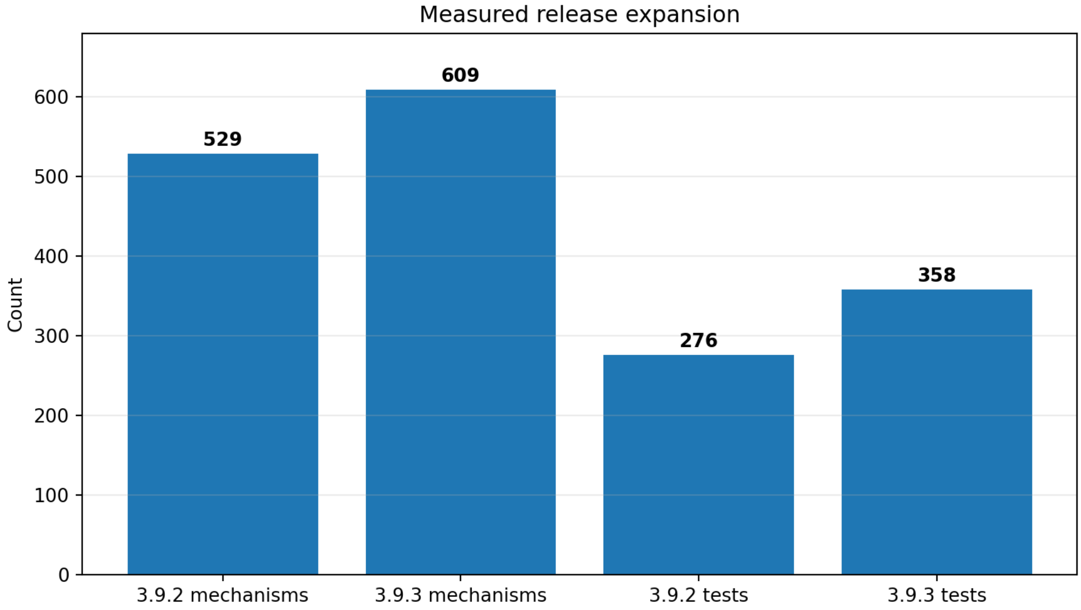

Figure 2. Mechanism and test counts before and after the IQ Field delta.

1. Release Decision and Source Survey

The public article inventory spans useful functions, procedural content, rendering, lighting, mathematics, fractals, compression, coding techniques, effects and presentations. Version 3.9.3 applies a strict admission rule: an idea enters the formal core only when it can be represented as a typed finite operator, exact or bounded field capability, finite query with termination status, deterministic procedural record, measurable filtering/color transform, execution contract or governance record.

SOURCE ACCESS NOTE

Live site blocked automated retrieval; survey used public index mirrors/update chronology and captured target URLs. The package does not redistribute article text or code. It records article titles, public target URLs, category, release disposition and a normalized UGTS mapping.

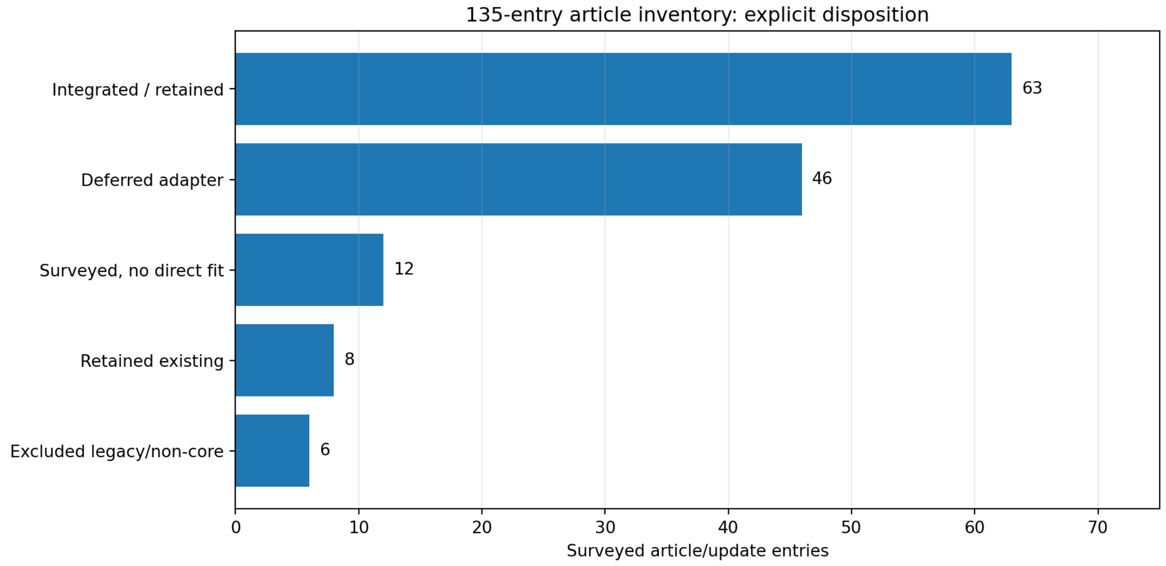

Figure 3. All 135 surveyed entries receive an explicit integrate, retain, defer, no-fit or exclude disposition.

Inventory coverage

Inventory category	Entries
useful maths	19
coding tricks	17
rendering techniques	16
recent additions	14
procedural content	13
fractal rendering	12
lighting	9
simple effects	8
fractals and dynamics	6
presentations	6
useful functions	6
compression	4
coding hacks	3
explorations	2

High-value admissions

• Useful remapping functions: normalized clamps, smooth transitions, impulses, gain and compact pulses.

• 2D and 3D distance functions: exact primitive relations, conservative bounds and explicit exactness capability.

• Intersection functions and SDF bounding: analytic ray intervals before expensive field evaluation.

• Raymarching SDFs and numerical normals: bounded tracing, terminal reason codes and finite gradient estimates.

• fBM, noise derivatives, Voronoi/Voronoise and domain warping: seeded finite procedural state with derivative and lineage contracts.

• Filterable procedurals and checker filtering: footprint-aware presentation functions that do not silently alter authoritative state.

• GPU conditionals, biplanar/triplanar mapping, gamma and color helpers: explicit execution and color-space policy.

• Recent smooth-minimum, rounded-box, XOR and domain-repetition updates: admitted with distance/exactness and identity boundaries.

Representative survey ledger

Year	Article/update	Category	Disposition
2025	GPU conditionals	recent additions	integrated-3.9.3-or-retained
2024	Smooth rounded boxes	recent additions	integrated-3.9.3-or-retained
2024	A study on smooth-minimums	recent additions	integrated-3.9.3-or-retained
2023	XOR of two SDFs	recent additions	integrated-3.9.3-or-retained
2023	Domain Repetition	recent additions	integrated-3.9.3-or-retained
2017	Intersection functions	useful functions	integrated-3.9.3-or-retained
2013	2D SDF functions	useful functions	integrated-3.9.3-or-retained
2008	3D SDF functions	useful functions	integrated-3.9.3-or-retained
2011	Useful remapping functions	useful functions	integrated-3.9.3-or-retained
2019	fBM	procedural content	integrated-3.9.3-or-retained
2017	Filterable procedurals	procedural content	integrated-3.9.3-or-retained
2017	Gradient noise derivatives	procedural content	integrated-3.9.3-or-retained
2008	Raymarching SDFs	procedural content	integrated-3.9.3-or-retained
2019	SDF Bounding Volumes	rendering techniques	integrated-3.9.3-or-retained
2015	Numerical normals for SDFs	rendering techniques	integrated-3.9.3-or-retained
2021	Biplanar mapping	recent additions	integrated-3.9.3-or-retained
2002	Domain warping	procedural content	integrated-3.9.3-or-retained

2. Formal 3.9.3 Architecture and State Contract

The retained UGTS authority chain is unchanged. A field or procedural query may produce a residual, interval, candidate hit, normal, material tag or presentation color. Discrete state changes only after support, compatibility, guard classification, verified proposal, deterministic transition and lineage/replay recording.

```
field/pattern + typed query
 -> conventional analytic bound
 -> local support
 -> compatibility
 -> relation residual + exactness/error status
 -> guard classification
 -> verified proposal
 -> deterministic transition
 -> lineage + replay/novelty log
 -> optional render/export
```

Extended substrate tuple

```
UGTS-KC 3.9.3 = UGTS-KC 3.9.2 + (Q, F, B, X, R, N, P, C, A)

Q : scalar remapping and analytic filtering
F : exact SDF primitives and generic field records
B : conservative distance and bounding capabilities
X : CSG, repetition and domain transforms
R : analytic intersections and bounded sphere tracing
N : deterministic noise, fBM, Voronoi and domain warping
P : projection, texturing and color helpers
C : GPU conditional and execution policy
A : article survey, admission ledger and evidence boundary
```

Minimum distance record

```
D = (distance, capability, material, source, error_bound, lineage)
capability in {exact_sdf, conservative_bound, implicit_residual}
```

Capability	Meaning	Permitted use
exact_sdf	Euclidean signed distance within the declared primitive/transform contract.	Direct distance marching, guards or collision only within the stated domain.
conservative_bound	Lower/upper safety estimate with declared direction and error semantics.	Pruning, bounded stepping or interval certification under a safety policy.
implicit_residual	Sign or scalar relation without Euclidean-distance status.	Root/guard classification with a suitable solver; never silently treated as distance.

NON-NEGOTIABLE AUTHORITY BOUNDARY

A raymarched pixel, filtered checker value, color palette sample, numerical normal or procedural noise value is downstream presentation unless an application separately promotes a certified relation under an explicit event contract.

3. Scalar Functions, Filtering and Color

The scalar layer provides small, inspectable functions that can be shared between CPU and shader profiles. Parameter order, finite domains and singular cases are checked instead of relying on undefined shader behavior.

Family	Reference relation	Contract
clamp/saturate	clamp(x,a,b); saturate(x)=clamp(x,0,1)	Reject reversed bounds where the API requires ordered limits.
smoothstep	t^2(3-2t)	C1 transition after clamped normalization.
smootherstep	t^3(6t^2-15t+10)	C2 transition for derivative-sensitive procedural signals.
inverse smoothstep	analytic inverse of cubic Hermite	Bounded to the unit interval and numerically guarded.
impulse/gain/power	finite remapping family	Shape controls are typed; negative or singular exponents reject.
cubic pulse	compact support around a center	Width must be positive; zero outside support.
sinc filter	sin(x)/x with stable origin limit	Taylor branch near zero avoids division instability.
filtered cosine/checker	analytic footprint average	Footprint belongs to presentation/error control, not identity.
cosine palette	a + b*cos(2*pi*(c*t+d))	Color generator only; component ranges are clamped at output.

Filtering rule

Version 3.9.3 distinguishes sampling a procedural function from integrating it over a finite pixel or query footprint. Analytic checker and periodic filters reduce aliasing without claiming perfect reconstruction for arbitrary signals. Event time and state identity remain unfiltered unless a separate filtered guard profile declares an error bound.

Color-space rule

```
linear = srgb_to_linear(encoded)
encoded = linear_to_srgb(linear)
weighted_average = linear_to_srgb(sum(w_i * srgb_to_linear(c_i)) / sum(w_i))
```

COLOR BOUNDARY

Gamma-correct filtering is a presentation requirement. It does not promote display RGB values to authoritative geometry, energy, material identity or legal provenance.

4. Signed-Distance Primitives and Capability Classes

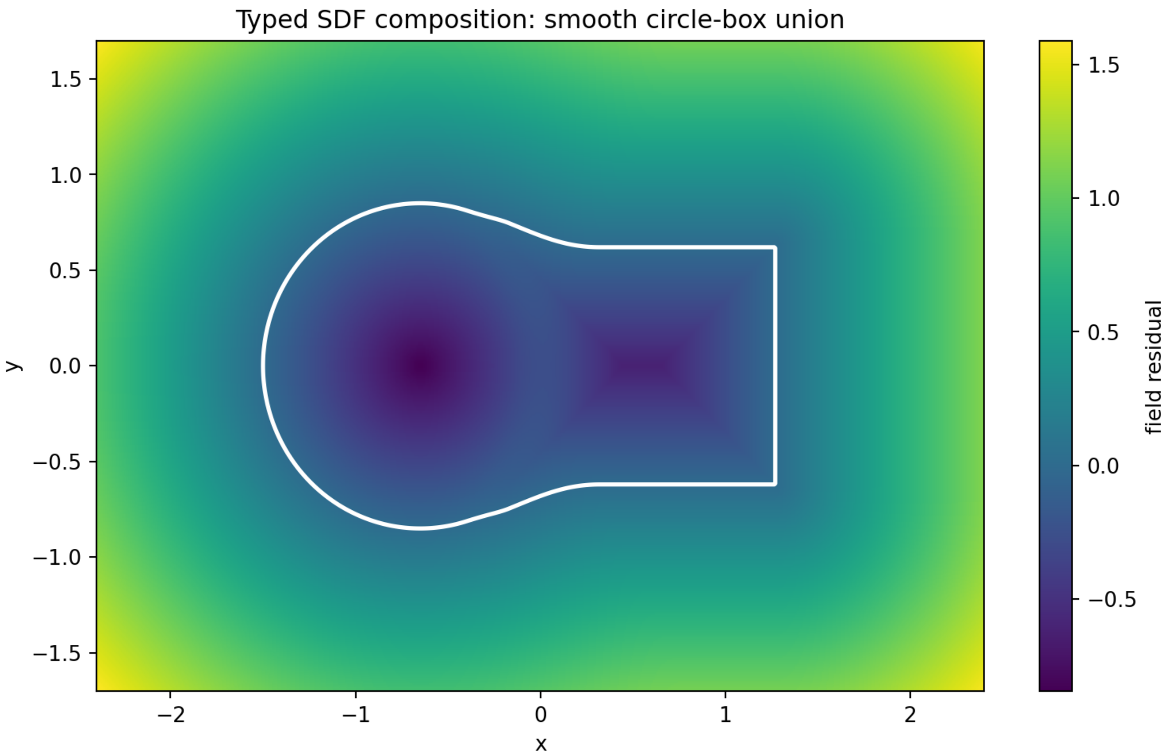

Figure 4. Smoothly composed 2D fields retain a zero relation, but smooth operators may lose exact Euclidean-distance status.

Class	Implemented operators	Capability
2D exact	circle, axis-aligned box, rounded box, segment, oriented box, triangle, regular polygon	Exact within declared finite parameters.
2D iterative	ellipse closest-distance magnitude	Exact sign; bounded iterative magnitude and terminal iteration contract.
3D exact	sphere, box, rounded box, plane, capsule, torus, capped cylinder, capped cone	Exact for the declared primitive formulas.
3D bound	ellipsoid analytic estimate	Conservative estimate except in exact reduction cases.

Primitive guards

- All sizes, radii and half-extents must be finite and positive where required.
- Triangle and polygon definitions reject degenerate orientation or unsupported side counts.
- Plane normals are normalized or rejected; the relation is not evaluated with a zero normal.
- The ellipse solver carries a finite iteration budget and does not claim a universal closed form.
- The ellipsoid estimate is labeled as a bound rather than promoted to exact distance.

EXACTNESS RULE

Exactness is a capability attached to a record, not inferred from a function name, a visually smooth contour or the presence of the letters SDF. Any transform or blend that invalidates the distance metric must downgrade the capability or supply a new bound.

5. CSG, Smooth Operators, Repetition and Transforms

Boolean composition is defined over sign-compatible fields. Exact primitive inputs preserve correct inside/outside semantics; exact distance quality after composition depends on the operator and location.

```
union(a,b)        = min(a,b)
intersection(a,b) = max(a,b)
difference(a,b)   = max(a,-b)
xor(a,b)          = max(min(a,b), -max(a,b))
```

Operator	Output contract	Authority note
hard union/intersection/difference	Correct set sign; piecewise distance behavior.	Topology and material selection remain explicit records.
exact XOR relation	Inside exactly one operand.	Does not imply a collision or event without compatibility and guard policy.
polynomial smooth union	Blended approximate field plus blend weight.	Useful for material interpolation; downgrade exactness.
smooth intersection/difference	Continuous blended relation.	Topology changes are committed separately from interpolation.
exponential smooth minimum	Soft minimum with finite sharpness.	Stable log-sum-exp implementation; not universally distance preserving.
rounding/onion	Offset or shell relation.	Shell identity and sign conventions are versioned.

Domain identity

```
cell_i  = floor(p_i / period_i + 1/2)
local_i = p_i - cell_i * period_i
identity = (cell, seed_profile, branch, phase, lineage)
```

Centered, limited and mirrored repetition return both local coordinates and an integer cell address. Equal local coordinates in different cells are not the same state. Twist and bend transforms retain an explicit transform profile. Uniform scale transports exact distance by the scale factor; nonuniform scale returns only a conservative bound unless a stronger proof is supplied.

Transform	Distance policy	Identity policy
centered repetition	Evaluate source field in local cell coordinates.	Retain integer cell address.
limited repetition	Clamp cell range before local evaluation.	Retain clamped cell and boundary status.
mirrored repetition	Reflect alternate cells under declared parity.	Retain cell plus mirror parity.
twist/bend	Generic deformation; exactness usually downgraded.	Retain transform parameters and source lineage.
uniform scale	Exact transport: $d' = \|s\| d$.	Scale is part of definition identity.
nonuniform scale	Lower-bound/safety estimate only.	Error/safety policy is mandatory.

6. Intersections, Bounded Sphere Tracing and Shading Queries

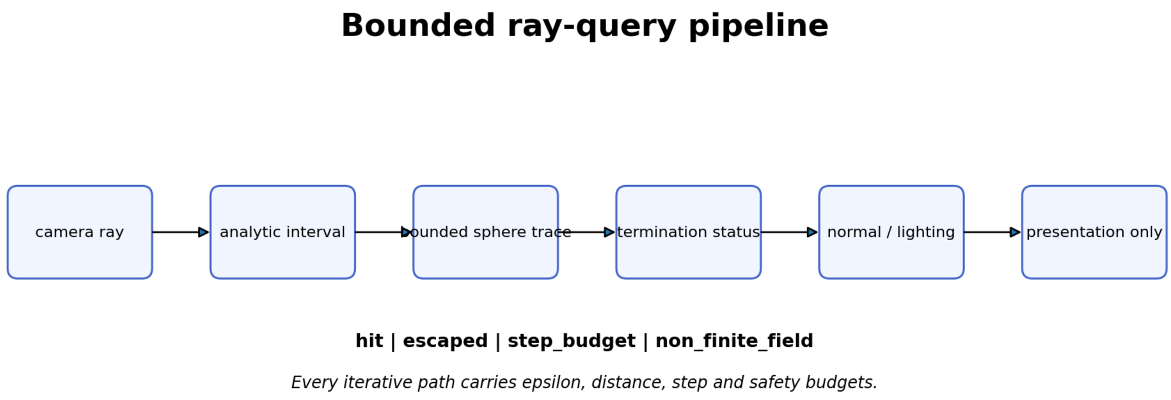

Figure 5. Analytic bounds and explicit termination make the ray-query profile finite and inspectable.

Query	Result	Failure/edge policy
ray-sphere	entry/exit interval	Reject zero direction; preserve tangent status.
ray-plane	single parameter t	Parallel or behind-origin results carry miss status.
ray-AABB	slab interval	Handles signed inverse directions and empty interval.
ray-triangle	t plus barycentric coordinates	Reject degenerate triangle and out-of-range barycentrics.
ray-disk	plane hit plus radius test	Normal and radius are validated.

Bounded sphere trace

```
input  = (ray, field, t_min, t_max, epsilon, max_steps, safety, min_step)
loop   = d <- field(ray(t))
         classify non-finite or hit
         t <- t + max(abs(d)*safety, min_step)
output = (hit, distance, position, residual, steps, status)
status in {hit, escaped, step_budget, non_finite_field}
```

A conservative sphere or AABB interval may narrow the march range. Exact SDFs may use a high safety factor; bounds and generic residuals require an application-specific Lipschitz/safety contract. A step-budget exit is never converted into a hit.

Finite shading queries

- Central and tetrahedral numerical normals carry an explicit epsilon.
- Soft shadow and ambient-occlusion probes use finite distance/sample budgets.
- Exponential fog is a typed transmittance function, not a physical atmosphere simulation.
- Perspective camera rays are generated from validated eye, target, up, field-of-view and aspect records.
- All lighting outputs remain presentation data unless separately promoted by an application contract.

7. Noise, fBM, Voronoi and Domain Warping

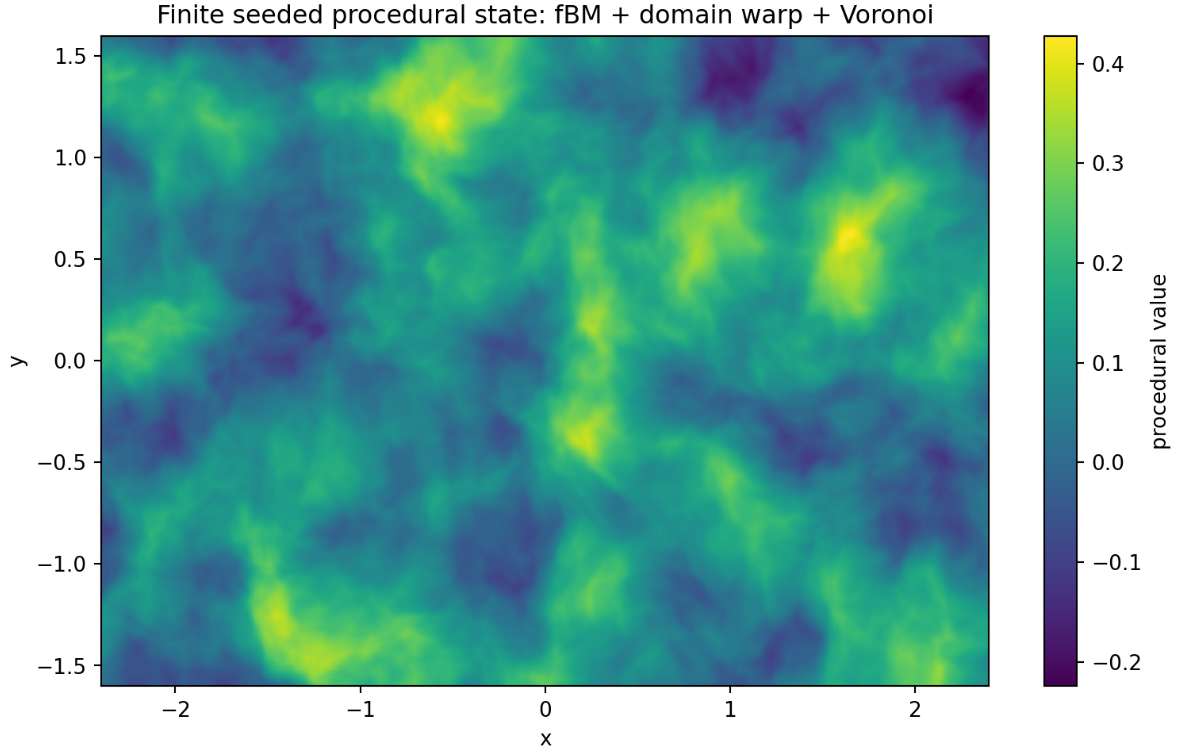

Figure 6. A deterministic seed, finite octave budget and retained warp/cell state define the procedural result.

The procedural layer uses fixed-width integer hashing and explicit seeds. Where supported, value and gradient noise return analytic derivatives. fBM transports derivatives through octave frequency scaling and stops at a declared octave count.

Operator	Returned state	Boundary
integer avalanche hash	deterministic uint32 value	Not cryptographic entropy or legal identity.
value noise	value plus analytic derivative	Finite lattice interpolation with explicit seed.
gradient noise	value plus analytic derivative	Gradient table/hash profile is versioned.
fBM	finite octave sum and derivative	Octave, gain and lacunarity budgets are validated.
turbulence/ridged	absolute/ridge transformed finite sum	Derivative discontinuities are documented.
domain warp	warped point plus warp vector	Source coordinates are not erased.
Voronoi	nearest, second, cell, feature point	Jitter and neighborhood radius are bounded.
smooth Voronoi/Voronoise	blended procedural value	Presentation field; not cell identity.

```
value = procedural(seed, cell, local_position, phase)
identity = (seed_profile, cell, branch, phase, lineage)
```

PROCEDURAL IDENTITY BOUNDARY

A sampled noise value is not identity, entropy proof or irreducible novelty. Stable identity retains the seed profile, cell address, branch/phase and lineage separately from the sampled scalar.

8. Texture Projection, GPU Policy and Android Integration

The projection layer adds triplanar and biplanar weights, repeated/offset UV transforms, sRGB/linear transfer and gamma-correct averaging. Biplanar mapping keeps the two strongest projection axes and renormalizes their weights; triplanar mapping remains available when three samples are justified.

Mechanism	Implementation	Boundary
triplanar weights	abs(normal)^exponent then normalize	Normal and exponent validated; texture samples are presentation.
biplanar top-two	select two dominant axes and renormalize	Reduces samples but can introduce profile-specific seams.
UV repeat/offset	fract/repeat under typed scale and offset	UV identity is not scene-node identity.
GPU conditional policy	branch, select or arithmetic mask profile	No architecture-independent speed claim.
min/max classification	ordered three-component extrema	Stable tie policy is explicit.

Android/GLES3 delta

- `android/UGTSKCKKijTGrove` is versioned to 3.9.3 while retaining the 3.9.2 Grove scene and mobile-control path.
- `iq_field.cpp/.hpp` provide C++20 scalar, SDF, repetition, noise and bounded sphere-trace helpers.
- `assets/shaders/iq_field_core.glsl` supplies reusable GLES field functions; the source tree also includes a WGSL counterpart.
- `grove_post.frag` uses derivative-footprint band-limited scan modulation and a branch-stable shock direction.
- The Android exporter template includes the C++/GLSL/profile assets and emits versionCode 393 / versionName 3.9.3.

GPU PERFORMANCE BOUNDARY

Branchless arithmetic is not assumed faster on every GPU. This release validates source structure and host/native consistency but does not claim a named physical-device speed, energy or thermal advantage. Promotion requires target compilation, equal-error comparison and measurement.

9. Reference Implementation, Demonstration and Validation

Module	Responsibility
ugts_kc3.iqmath	remapping, pulses, sinc/periodic filtering and cosine palettes
ugts_kc3.iqsdf	2D/3D primitives, capability records, CSG, smooth operators, repetition and transforms
ugts_kc3.iqray	analytic intersections, bounded tracing, normals, finite shadow/AO, fog and camera rays
ugts_kc3.iqnoise	deterministic hashing, noise derivatives, finite fBM, domain warping and Voronoi
ugts_kc3.iqtexture	triplanar/biplanar projection, UV transforms and color-space operations
ugts_kc3.iqdemo	dependency-free deterministic PPM demonstration and metrics
native/Android iq_field	C++20 and GLSL ES integration subset

Deterministic IQ Field Grove demonstration

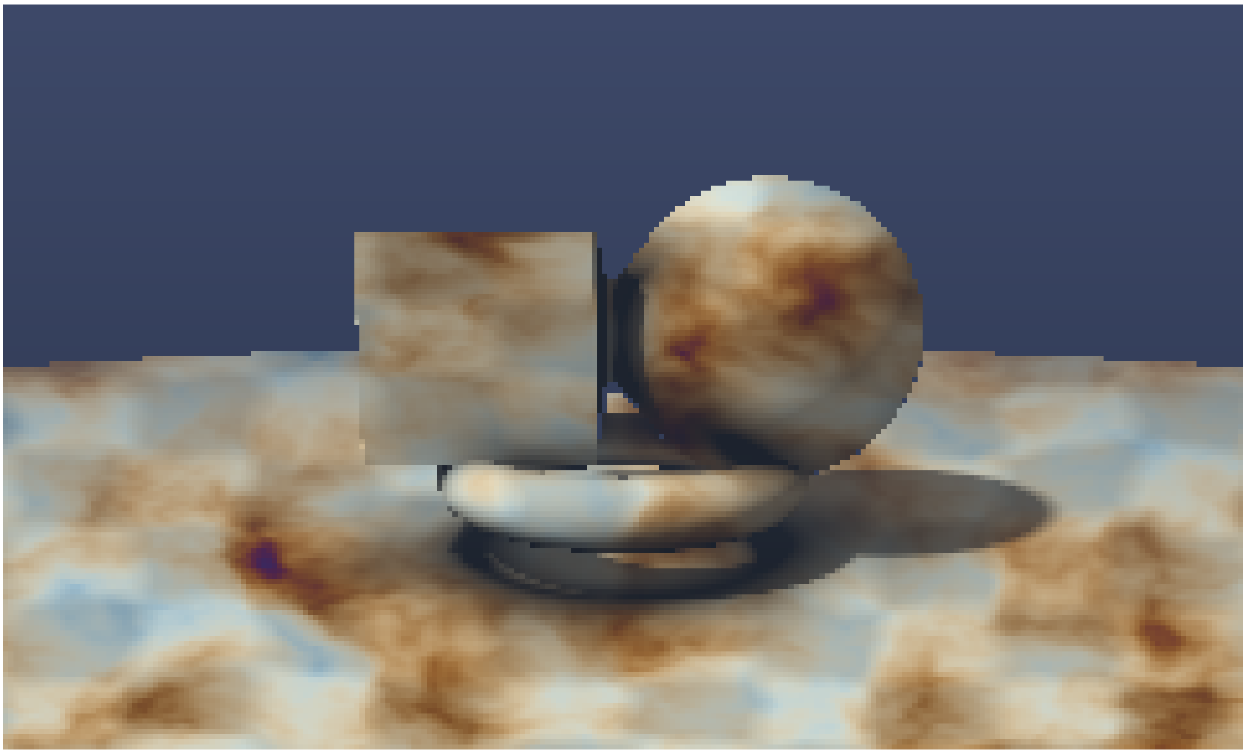

Figure 7. The included 240 x 144 reference render exercises SDF composition, bounded tracing, tetrahedral normals, finite shadow/AO, fog, fBM, domain warping, Voronoi and analytic checker filtering.

Metric	Captured value
Resolution / pixels	240 x 144 / 34560
Hit count / ratio	21063 / 0.609462
Total / mean trace steps	722398 / 20.902720
Deterministic seed	393
Output	PPM + PNG + summary JSON + README

Validation gates

Gate	Result	Established scope
Python compile	PASS	All source modules and tests compile.
Automated tests	358/358 PASS	276 retained + 82 new IQ tests.
C++20 host smoke	PASS	Native IQ field helper compiles and returns expected SDF/repetition/trace results.
JSON/Schema	PASS	79 JSON files parse; IQ profile has zero schema errors.

Gate	Result	Established scope
Catalog	PASS	Engineering IDs are contiguous M198-M609; delta M530-M609 contains 80 rows.
Android export	PASS	Generated project contains scene pack, version 3.9.3 metadata and IQ assets.
Source distribution	PASS	Versioned Python sdist generated under dist/.
Physical Android/GPU benchmark	NOT CLAIMED	Requires named device, driver, build, power/thermal state and equal-error oracle.

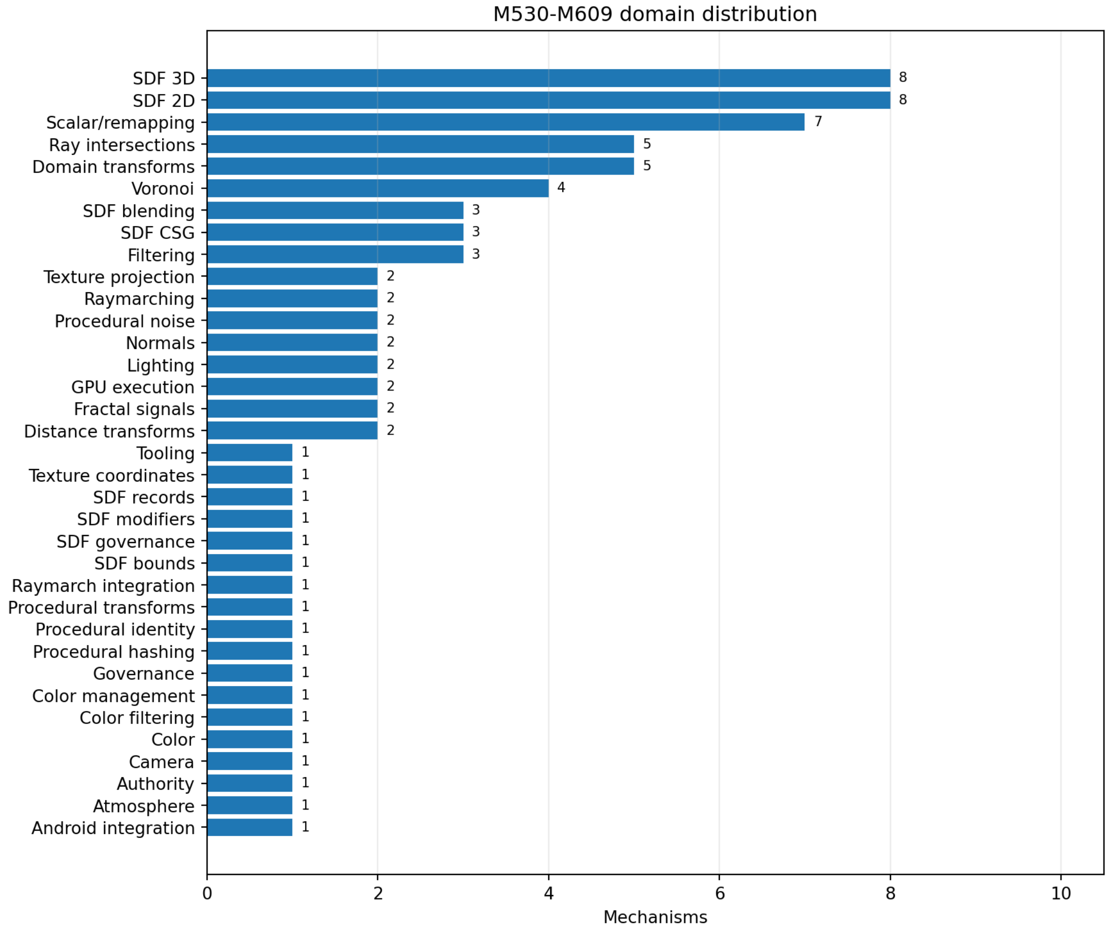

Figure 8. The 80-mechanism delta is distributed across field, query, procedural, filtering, projection, integration and governance domains.

10. Complete Mechanism Catalog M530-M609

The following 80 entries are engineering additions. They extend the retained source-normalized and engineering catalog without renumbering earlier mechanisms. Validation labels refer only to the shipped reference files and tests.

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M530	Scalar/remapping	Saturate and ordered clamp	**Definition:** Finite interval projection with explicit reversed-bound rejection. **Formula:** clamp(x,a,b); saturate(x)=clamp(x,0,1) **Integration:** Shared CPU/shader guard helper. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M531	Scalar/remapping	Cubic smoothstep	**Definition:** C1 Hermite transition with clamped input. **Formula:** t^2(3-2t) **Integration:** Animation, filtering and guard ramps; never replaces event classification. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M532	Scalar/remapping	Quintic smootherstep	**Definition:** C2 transition for derivative-sensitive procedural signals. **Formula:** t^3(6t^2-15t+10) **Integration:** Noise interpolation and presentation fades. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M533	Scalar/remapping	Inverse smoothstep	**Definition:** Closed inverse on the clamped unit interval. **Formula:** 0.5-sin(asin(1-2x)/3) **Integration:** Parameter reconstruction and authoring tools. **Source family:** Inverse smoothstep **Validation:** Implemented + tested
M534	Scalar/remapping	Impulse and exponential step family	**Definition:** Bounded peak and monotone decay operators with typed positive shape parameters. **Formula:** k*x*exp(1-k*x); exp(-k*x^n) **Integration:** Gameplay envelopes and procedural modulation. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M535	Scalar/remapping	Gain, parabola and power curves	**Definition:** Normalized unit-interval shaping functions. **Formula:** gain(x,k); [4x(1-x)]^k; normalized x^a(1-x)^b **Integration:** Color, timing and distribution shaping. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M536	Scalar/remapping	Compact cubic pulse	**Definition:** Finite-support smooth pulse around a center and width. **Formula:** 1-smoothstep(|x-c|/w) **Integration:** Impact flashes, field bands and UI emphasis. **Source family:** Useful remapping functions **Validation:** Implemented + tested
M537	Filtering	Sinc and band-limited cosine	**Definition:** Analytic box average of a cosine over a finite pixel footprint. **Formula:** cos(x)*sin(w)/w **Integration:** Anti-aliasing for procedural periodic signals. **Source family:** Band-limiting procedural signals **Validation:** Implemented + tested
M538	Filtering	Exact 1D checker box filter	**Definition:** Exact antiderivative-based average of a period-two binary checker. **Formula:** [I(x+w)-I(x-w)]/(2w) **Integration:** Stable procedural checker LOD without texture samples. **Source family:** Analytic checker filtering **Validation:** Implemented + tested
M539	Color	Cosine palette	**Definition:** Compact periodic RGB palette with four three-vectors. **Formula:** a+b*cos(2*pi*(c*t+d)) **Integration:** Procedural materials and deterministic demo coloring. **Source family:** Simple color palettes **Validation:** Implemented + tested
M540	SDF 2D	Circle signed distance	**Definition:** Exact Euclidean signed distance to a circle. **Formula:** length(p)-r **Integration:** 2D vector/collision/field authoring. **Source family:** 2D SDF functions **Validation:** Implemented + tested
M541	SDF 2D	Axis-aligned box signed distance	**Definition:** Exact signed distance to a rectangle centered at the origin. **Formula:** length(max(abs(p)-b,0))+min(max(q.x,q.y),0) **Integration:** 2D field composition and guards. **Source family:** 2D SDF functions **Validation:** Implemented + tested
M542	SDF 2D	Rounded-box signed distance	**Definition:** Box distance over contracted extents followed by a radius offset. **Formula:** sdBox(p,max(b-r,0))-r **Integration:** Smooth UI and collision proxy authoring. **Source family:** 2D SDF functions / smooth rounded boxes **Validation:** Implemented + tested
M543	SDF 2D	Line-segment distance	**Definition:** Exact point-to-segment projection with a degenerate-endpoint path. **Formula:** length(pa-ba*clamp(dot(pa,ba)/dot(ba,ba),0,1)) **Integration:** Trajectory, stroke and capsule construction. **Source family:** 2D SDF functions / line segment SDF **Validation:** Implemented + tested

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M544	SDF 2D	Oriented box distance	**Definition:** Transform to a segment-aligned frame and evaluate a box. **Formula:** sdBox(R^T(p-mid),(length/2,thickness)) **Integration:** Oriented selection and road/beam profiles. **Source family:** 2D SDF functions **Validation:** Implemented + tested
M545	SDF 2D	Triangle signed distance	**Definition:** Exact edge-projection distance with an orientation sign. **Formula:** min edge distance with oriented half-space sign **Integration:** 2D collision and procedural masks. **Source family:** Distance to triangle / 2D SDF functions **Validation:** Implemented + tested
M546	SDF 2D	Regular polygon distance	**Definition:** Fold polar angle into a side sector and evaluate the supporting line. **Formula:** cos(localAngle)*length(p)-r **Integration:** Bounded n-gon procedural field primitive. **Source family:** 2D SDF functions **Validation:** Implemented + tested
M547	SDF 2D	Ellipse closest-distance solver	**Definition:** Newton closest-point magnitude plus exact implicit sign; iteration budget is explicit. **Formula:** min_theta ||(a cos theta,b sin theta)-p|| **Integration:** Bounded numerical path; not silently advertised as a closed exact formula. **Source family:** Distance to an ellipse / working with ellipses **Validation:** Implemented + bounded tests
M548	SDF 3D	Sphere signed distance	**Definition:** Exact Euclidean signed distance to a sphere. **Formula:** length(p)-r **Integration:** Ray queries, collision and procedural geometry. **Source family:** 3D SDF functions / sphere functions **Validation:** Implemented + tested
M549	SDF 3D	Axis-aligned box signed distance	**Definition:** Exact signed distance to a centered box. **Formula:** length(max(abs(p)-b,0))+min(max(q),0) **Integration:** 3D field composition and bounding. **Source family:** 3D SDF functions / box functions **Validation:** Implemented + tested
M550	SDF 3D	Rounded-box signed distance	**Definition:** Exact rounded-box construction for uniform radius. **Formula:** sdBox(p,max(b-r,0))-r **Integration:** Mobile-friendly procedural solids. **Source family:** 3D SDF functions / smooth rounded boxes **Validation:** Implemented + tested
M551	SDF 3D	Plane signed relation	**Definition:** Normalized plane residual with explicit normal validation. **Formula:** dot(normalize(n),p)+h **Integration:** Ground, clipping and event surfaces. **Source family:** 3D SDF functions / intersection functions **Validation:** Implemented + tested
M552	SDF 3D	Capsule signed distance	**Definition:** Exact distance to a line segment swept by a sphere. **Formula:** distance(p,segment(a,b))-r **Integration:** Characters, beams and sweep guards. **Source family:** 3D SDF functions **Validation:** Implemented + tested
M553	SDF 3D	Torus signed distance	**Definition:** Exact distance to a standard y-axis torus. **Formula:** length((length(p.xz)-R,p.y))-r **Integration:** Procedural scene and topology visualization. **Source family:** 3D SDF functions **Validation:** Implemented + tested
M554	SDF 3D	Capped cylinder signed distance	**Definition:** Exact radial/axial composition for a finite y-axis cylinder. **Formula:** min(max(d),0)+length(max(d,0)) **Integration:** Columns, triggers and collision proxies. **Source family:** 3D SDF functions / disk-cylinder bounds **Validation:** Implemented + tested
M555	SDF 3D	Capped cone signed distance	**Definition:** Exact meridian-segment/cap construction for a finite frustum. **Formula:** signed min of cap and side-segment distances **Integration:** Cone/frustum support and procedural models. **Source family:** 3D SDF functions / intersection functions **Validation:** Implemented + tested
M556	SDF bounds	Ellipsoid distance bound	**Definition:** Conservative analytic estimate exact for spheres and principal axes. **Formula:** k0*(k0-1)/k1 **Integration:** Typed as a bound so sphere tracing applies safety and validation. **Source family:** Ellipsoids / distance estimation to implicits **Validation:** Implemented + bounded tests
M557	SDF CSG	Union and intersection	**Definition:** Minimum and maximum preserve signed inside/outside semantics for exact inputs. **Formula:** union=min(a,b); intersection=max(a,b) **Integration:** Canonical field algebra. **Source family:** 3D SDF functions **Validation:** Implemented + tested
M558	SDF CSG	Difference	**Definition:** Subtract B from A using sign inversion and intersection. **Formula:** max(a,-b) **Integration:** Destructive field authoring with explicit topology status. **Source family:** 3D SDF functions **Validation:** Implemented + tested
M559	SDF CSG	Exact XOR of two SDFs	**Definition:** Symmetric difference field using nested min/max signs. **Formula:** max(min(a,b),-max(a,b)) **Integration:** First-class exclusive region and event guard. **Source family:** XOR of two SDFs **Validation:** Implemented + tested

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M560	SDF blending	Polynomial smooth union with blend weight	**Definition:** Compact finite-radius smooth minimum returning both distance and ownership weight. **Formula:** mix(b,a,h)-k*h*(1-h) **Integration:** Material blending and smooth scene composition. **Source family:** Smooth minimum / study on smooth-minimums **Validation:** Implemented + tested
M561	SDF blending	Smooth intersection and difference	**Definition:** Sign-dual polynomial operations with the same radius contract. **Formula:** -smin(-a,-b,k); -smin(-a,b,k) **Integration:** Complete smooth CSG family. **Source family:** Smooth minimum / study on smooth-minimums **Validation:** Implemented + tested
M562	SDF blending	Exponential smooth minimum	**Definition:** Log-sum-exp smooth minimum with explicit positive sharpness. **Formula:** -log(exp(-ka)+exp(-kb))/k **Integration:** Alternative blend with unbounded influence disclosed. **Source family:** Smooth minimum / study on smooth-minimums **Validation:** Implemented + tested
M563	SDF modifiers	Rounding and onion shells	**Definition:** Distance offsets create rounded dilation and finite-thickness shells. **Formula:** d-r; abs(d)-t **Integration:** Field morphology with topology change tracked separately. **Source family:** 3D SDF functions / interior SDF modeling **Validation:** Implemented + tested
M564	SDF records	Distance/material/source record	**Definition:** A finite record carries distance, exactness, material and source identity. **Formula:** (d, exact, material, source) **Integration:** Prevents scalar distance from erasing provenance and promotion status. **Source family:** Distance functions / UGTS referential integration **Validation:** Implemented + tested
M565	SDF governance	Exactness capability boundary	**Definition:** Exact SDF, conservative bound and generic implicit residual are distinct declared capabilities. **Formula:** capability in {exact,bound,residual} **Integration:** Required before marching, collision or event promotion. **Source family:** Distance functions / SDF bounding volumes **Validation:** Implemented + tested
M566	Domain transforms	Centered domain repetition with cell identity	**Definition:** Center each period around zero while retaining integer cell coordinates. **Formula:** cell=floor(p/c+0.5); local=p-cell*c **Integration:** Procedural instancing without collapsing lineage to local coordinates. **Source family:** Domain Repetition **Validation:** Implemented + tested
M567	Domain transforms	Bounded domain repetition	**Definition:** Clamp repetition cell indices to declared lower/upper bounds. **Formula:** cell=clamp(round(p/c),lo,hi) **Integration:** Finite support and predictable work. **Source family:** Domain Repetition **Validation:** Implemented + tested
M568	Domain transforms	Mirrored repetition	**Definition:** Flip local coordinates on odd cells while preserving cell ID. **Formula:** local*=(-1)^cell **Integration:** Seam-reduced repeated motifs and chirality metadata. **Source family:** Domain Repetition **Validation:** Implemented + tested
M569	Domain transforms	Y-axis twist transform	**Definition:** Rotate xz by an angle proportional to y. **Formula:** R(rate*y)*p.xz **Integration:** Procedural deformation; distance status downgraded unless certified. **Source family:** 3D SDF functions / plane deformations **Validation:** Implemented + tested
M570	Domain transforms	X-directed bend transform	**Definition:** Map x to a circular arc with zero-curvature identity path. **Formula:** angle=k*x; radius=1/k **Integration:** Procedural deformation with a bounded singularity guard. **Source family:** 3D SDF functions / plane deformations **Validation:** Implemented + tested
M571	Distance transforms	Uniform-scale exact distance transport	**Definition:** Inverse-transform the query and multiply distance by absolute scale. **Formula:** d_world=|s|*d_local(p/s) **Integration:** Exact distance under uniform scale. **Source family:** Distance functions **Validation:** Implemented + tested
M572	Distance transforms	Nonuniform-scale lower bound	**Definition:** Multiply local distance by the smallest absolute axis scale. **Formula:** d_bound=min(|s|)*d_local(p/s) **Integration:** Conservative marcher step; not exact distance. **Source family:** Distance estimation to implicits **Validation:** Implemented + tested
M573	Ray intersections	Ray-sphere interval	**Definition:** Stable quadratic reduction for normalized rays. **Formula:** t=-b +/- sqrt(b^2-c) **Integration:** Bounding volumes and analytic hit tests. **Source family:** Intersection functions / sphere functions **Validation:** Implemented + tested
M574	Ray intersections	Ray-plane intersection	**Definition:** Parallel rejection plus non-negative distance policy. **Formula:** t=-(dot(n,o)+h)/dot(n,d) **Integration:** Portal, ground and clipping queries. **Source family:** Intersection functions **Validation:** Implemented + tested
M575	Ray intersections	Ray-AABB slab interval	**Definition:** Per-axis near/far intersection with parallel-axis handling. **Formula:** near=max(tmin_i); far=min(tmax_i) **Integration:** SDF bounding volumes and BVH integration. **Source family:** Intersection functions / box functions **Validation:** Implemented + tested

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M576	Ray intersections	Ray-triangle barycentric hit	**Definition:** Moller-Trumbore hit with optional backface culling. **Formula:** (t,u,v) from edge determinant **Integration:** Mesh/SDF hybrid reference queries. **Source family:** Intersection functions **Validation:** Implemented + tested
M577	Ray intersections	Ray-disk intersection	**Definition:** Plane hit followed by finite radial admission. **Formula:** plane_t and ||hit-center||<=r **Integration:** Disc/cylinder support and analytic shadows. **Source family:** Intersection functions / disk-cylinder bounds **Validation:** Implemented + tested
M578	Raymarching	Bounded sphere tracer	**Definition:** Epsilon, distance range, step, safety and iteration budgets are explicit. **Formula:** t += max(|d|*safety,min_step) **Integration:** Deterministic CPU oracle and shader parity target. **Source family:** Raymarching SDFs **Validation:** Implemented + tested
M579	Raymarching	Trace termination certificate	**Definition:** Return hit/escaped/budget/non-finite status, residual, distance and step count. **Formula:** TraceResult(hit,t,p,residual,steps,status) **Integration:** No silent infinite loop or false exactness claim. **Source family:** Raymarching SDFs / SDF bounding volumes **Validation:** Implemented + tested
M580	Normals	Central-difference field normal	**Definition:** Six-sample normalized gradient estimate. **Formula:** grad=(f(p+ex)-f(p-ex),...) **Integration:** Reference quality path. **Source family:** Numerical normals for SDFs **Validation:** Implemented + tested
M581	Normals	Tetrahedral field normal	**Definition:** Four-sample tetrahedral gradient estimate. **Formula:** sum e_i*f(p+eps*e_i) **Integration:** Lower sample-count mobile path. **Source family:** Numerical normals for SDFs **Validation:** Implemented + tested
M582	Lighting	SDF soft shadow	**Definition:** Secondary march tracks the minimum distance-to-ray ratio. **Formula:** visibility=min(visibility,k*h/t) **Integration:** Bounded presentation-only shadow estimate. **Source family:** Sphere soft shadow / penumbra shadows **Validation:** Implemented + tested
M583	Lighting	SDF ambient occlusion	**Definition:** Sample field along the surface normal with decaying weights. **Formula:** AO=1-sum(w*max(0,s-d))/normalizer **Integration:** Presentation descriptor; not an authority signal. **Source family:** Sphere ambient occlusion / multires ambient occlusion **Validation:** Implemented + tested
M584	Atmosphere	Exponential fog transmittance	**Definition:** Distance/density extinction with configurable exponent. **Formula:** exp(-(density*distance)^n) **Integration:** Deterministic mobile post-lighting effect. **Source family:** Better fog **Validation:** Implemented + tested
M585	Camera	Perspective camera-ray construction	**Definition:** Orthonormal basis and tangent FOV projection from normalized UV. **Formula:** d=normalize(f+u*r+v*up) **Integration:** Raymarch preview and test harness. **Source family:** Basic raytracer / avoiding trigonometry **Validation:** Implemented + tested
M586	Raymarch integration	AABB/sphere bound before field march	**Definition:** Analytic bounds restrict the ray interval before iterative SDF evaluation. **Formula:** [t0,t1]=intersect(bound); march only inside **Integration:** Connects retained BVH/spatial stage to IQ fields. **Source family:** SDF Bounding Volumes **Validation:** Implemented + tested
M587	Authority	Render-field downstream boundary	**Definition:** Raymarch hits, normals, AO and color remain presentation unless an application promotes a certified relation. **Formula:** field -> bounded query -> optional render **Integration:** Preserves UGTS support/compatibility/guard/event authority. **Source family:** Raymarching SDFs + UGTS canonical sequence **Validation:** Implemented + tested
M588	Procedural hashing	Integer avalanche hash	**Definition:** Portable 32-bit hash with fixed overflow semantics. **Formula:** xor-shift/multiply/xor-shift **Integration:** Deterministic lattice seeds without sine-based undefined parity. **Source family:** Float and random **Validation:** Implemented + tested
M589	Procedural noise	Value noise with analytic derivatives	**Definition:** Quintic interpolation and exact derivative of the interpolant. **Formula:** bilinear values with u=f5(f); d/df **Integration:** Terrain, animation and field perturbation. **Source family:** Value noise derivatives **Validation:** Implemented + tested
M590	Procedural noise	Gradient noise with analytic derivatives	**Definition:** Lattice gradient dots and differentiated quintic interpolation. **Formula:** noise(p), grad noise(p) **Integration:** Normal-aware procedural content. **Source family:** Gradient noise derivatives **Validation:** Implemented + tested
M591	Fractal signals	Finite fBM with derivative transport	**Definition:** Budgeted octave sum with declared lacunarity, gain and seed schedule. **Formula:** sum a_i noise(f_i p); grad includes f_i **Integration:** Procedural terrain/material signal. **Source family:** fBM **Validation:** Implemented + tested

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M592	Fractal signals	Turbulence and ridged fBM	**Definition:** Absolute and inverted-absolute finite octave sums. **Formula:** sum a_i|n_i|; sum a_i(1-|n_i|)^2 **Integration:** Cloud/ridge appearance with finite budgets. **Source family:** fBM / filterable procedurals **Validation:** Implemented + tested
M593	Procedural transforms	Two-channel domain warping	**Definition:** Offset coordinates by two seeded fBM channels and return the warp vector. **Formula:** p'=p+s(qx,qy) **Integration:** Procedural structure with source coordinates retained. **Source family:** Domain warping **Validation:** Implemented + tested
M594	Voronoi	Nearest/second feature query	**Definition:** Search a bounded 3x3 lattice neighborhood and return distances, cell and feature point. **Formula:** F1,F2,cell,feature **Integration:** Cell identity, placement and material regions. **Source family:** Voronoi / Voronoi edges **Validation:** Implemented + tested
M595	Voronoi	Voronoi edge estimate	**Definition:** Half the F2-F1 gap provides a local boundary estimate. **Formula:** 0.5*max(F2-F1,0) **Integration:** Cell-border masks and guards. **Source family:** Voronoi edges **Validation:** Implemented + tested
M596	Voronoi	Smooth Voronoi value field	**Definition:** Distance-weighted lattice values with positive sharpness. **Formula:** sum exp(-k*d_i)*v_i / sum weights **Integration:** Soft cellular materials. **Source family:** Smooth Voronoi **Validation:** Implemented + tested
M597	Voronoi	Voronoise	**Definition:** Random feature placement and compact-support value blending. **Formula:** sum w(d_i)*v_i / sum w(d_i) **Integration:** Continuous cellular/noise hybrid. **Source family:** Voronoise **Validation:** Implemented + tested
M598	Filtering	Analytic 2D checker filter	**Definition:** Combine independent exact 1D box averages into checker parity probability. **Formula:** ax*ay+(1-ax)(1-ay) **Integration:** Mobile procedural texture without texture fetches. **Source family:** Filterable procedurals / analytic checker filtering **Validation:** Implemented + tested
M599	Procedural identity	Seed and lineage separation	**Definition:** Seed schedules are deterministic inputs; cell/branch identity remains separate from sampled value. **Formula:** value=f(seed,cell,p); lineage=(seed,cell,phase) **Integration:** Prevents noise values from becoming state identity. **Source family:** fBM / UGTS lineage **Validation:** Implemented + tested
M600	Texture projection	Triplanar weights	**Definition:** Normalize powered absolute normal components across three projections. **Formula:** w=normalize(|n|^k) **Integration:** Textureless/procedural mapping and conventional texture adapters. **Source family:** Texturing and raymarching **Validation:** Implemented + tested
M601	Texture projection	Biplanar top-two projection	**Definition:** Select and blend the two dominant projection axes. **Formula:** axes=top2(|n|); w=normalize(|n_axes|^k) **Integration:** Two-sample projection policy with explicit axis output. **Source family:** Biplanar mapping **Validation:** Implemented + tested
M602	Texture coordinates	Repeat/offset UV transform	**Definition:** Finite repeat and offset followed by unit wrapping. **Formula:** uv'=fract(uv*repeat+offset) **Integration:** Conventional texture repetition downstream of authoritative geometry. **Source family:** Texture repetition **Validation:** Implemented + tested
M603	Color management	sRGB/linear transfer pair	**Definition:** Piecewise IEC-style transfer functions for scalar channels. **Formula:** linear=decode(sRGB); sRGB=encode(linear) **Integration:** Reproducible preview and export color. **Source family:** Gamma correct blurring **Validation:** Implemented + tested
M604	Color filtering	Gamma-correct weighted average	**Definition:** Decode samples to linear light, average, then re-encode. **Formula:** encode(sum w_i*decode(c_i)) **Integration:** Correct blur and mip-like reference operations. **Source family:** Gamma correct blurring **Validation:** Implemented + tested
M605	GPU execution	Typed conditional policy	**Definition:** Branching versus arithmetic select is an explicit profile choice, not a universal micro-optimization. **Formula:** select(predicate,a,b) with target policy metadata **Integration:** CPU/shader parity and mobile profiling. **Source family:** GPU conditionals **Validation:** Implemented + tested
M606	GPU execution	Three-component min/max classification	**Definition:** Return extrema and their component indices for branch/control use. **Formula:** min/max(x,y,z), argmin/argmax **Integration:** Color thresholds, normal-axis selection and diagnostics. **Source family:** The meaning of max(x,y,z) **Validation:** Implemented + tested

ID	Domain	Mechanism	Definition / formula / integration / source / validation
M607	Android integration	GLSL ES IQ field include and filtered Grove post process	**Definition:** Shader assets expose safe SDF/filter/color helpers and use analytic footprint modulation in the checked-in app. **Formula:** GLSL ES 3.00 field/filter functions **Integration:** Directly extends the 3.9.2 POCO/Grove source without replacing its scene runtime. **Source family:** GPU conditionals / band-limiting procedural signals **Validation:** Implemented + source validation
M608	Tooling	IQ Field CLI demo and profile schema	**Definition:** A bounded dependency-free demo emits a PPM, JSON metrics and release trace under a versioned schema. **Formula:** ugts-kc iq-demo OUTPUT **Integration:** Executable evidence for SDF marching and procedural signals. **Source family:** Raymarching SDFs / procedural content **Validation:** Implemented + smoke tested
M609	Governance	Article-survey admission and kill criteria	**Definition:** Only functionality with typed domains, finite budgets, exactness status and UGTS handoff enters the core; full path tracing, unbounded fractals, legacy 4K/SSE hacks and unsupported performance claims remain excluded or future work. **Formula:** survey -> admit/defer/reject -> tests -> manifest **Integration:** Makes source-derived inspiration visibly distinct from engineering implementation. **Source family:** Articles index / current update inventory **Validation:** Documented + machine checked

11. Compatibility, Boundaries, Kill Criteria and Package Map

Compatibility

• The retained `ugts_kc` and `ugts_kc3` modules remain available under their previous import paths.

• The game-project schema stays `ugts-kc-game-project-3.9`; the mobile 3D schema and KC3D392 scene-pack format remain readable.

• The IQ modules are namespaced and additive; similarly named legacy geometry functions are not overwritten.

• The Android native application keeps its GLES3 renderer, fixed-step game world, adaptive quality and Grove presentation while receiving optional IQ helpers.

• The 3.9.2 catalog repair adds only M529 `GroveReleaseClosure`; all prior IDs and semantics remain unchanged.

Explicit non-claims

• No complete path tracer, production ray tracer, unbounded fractal renderer, VR platform stack or compression suite is claimed.

• No article text, copyrighted code or third-party asset is redistributed by the survey.

• No generic implicit field is silently labeled an exact Euclidean signed distance.

• No universal convergence guarantee is claimed for sphere tracing arbitrary residuals or deformed fields.

• No physical Android/GPU performance, energy, thermal or cross-vendor bit-identity claim is promoted.

• Requester identity, attribution and ownership statements remain requester-supplied and unverified.

Kill criteria

• Distance exactness or a conservative bound cannot be established for the intended guard or step policy.

• Iteration, octave, cell-neighborhood, ray-step or sample budgets exceed the application latency/memory envelope.

• Repetition or procedural sampling discards cell, seed, branch, phase or lineage needed for identity reconstruction.

• Filtering, color conversion or numerical gradients alter event ordering beyond the declared margin.

• GPU conditional style, extra texture samples or field evaluation costs more than the retained conventional path at equal error.

• A full renderer/effect-specific article family would dominate the finite query-first core and is better kept as a downstream adapter.

CORRECTION RECORDED

The supplied 3.9.2 file labeled M450-M529 contained 79 rows and ended at M528. Version 3.9.3 restores M529 as `GroveReleaseClosure` without renumbering earlier entries. The correction is machine-readable in the combined catalog and source-basis record.

Package map

Path	Contents
src/ugts_kc3/iq*.py	Python scalar, SDF, ray, procedural, texture and demonstration modules.
spec/	Formal definition, M530-M609 catalog, combined M198-M609 catalog, schema, profile and 135-entry survey.
shaders/	GLSL ES and WGSL IQ Field helper sources.
android/UGTSKCKKijTGrove/	Checked-in 3.9.3 NativeActivity source with Grove and IQ integration.
src/ugts_kc3/android_template/	Versioned Android exporter template including IQ C++/GLSL/profile assets.
examples/iq_field_grove/	Deterministic PPM/PNG render, summary JSON and README.

Path	Contents
tests/test_ugts_kc393.py	82 new individually counted IQ Field tests.
validation/	Compile, tests, native smoke, schema/catalog/shader, Android export and distribution evidence.
report/	This PDF and retained legacy reports.
tools/	Figure and report generators for the 3.9.3 deliverable.

Reproduction

```
PYTHONPATH=src python -m unittest discover -s tests -v
PYTHONPATH=src python -m ugts_kc3 info
PYTHONPATH=src python -m ugts_kc3 iq-demo examples/iq_field_grove --width 240 --height 144
PYTHONPATH=src python -m ugts_kc3 build-android examples/grove_k_kij_t_3d/project.json out/android --profil
python setup.py sdist
```

Appendix A - Public Article Source Basis and Attribution

Survey method

The report used the public articles index, indexed mirrors/update chronology and captured public target URLs because direct automated retrieval of the live pages was blocked in the execution environment. Titles and categories were used for an engineering admission survey; no article body was reproduced. The full 135-entry ledger is included in JSON and CSV.

Source	Public target URL
Article inventory	https://iquilezles.org/articles/
Useful remapping functions	https://iquilezles.org/www/articles/functions/functions.htm
2D SDF functions	https://iquilezles.org/www/articles/distfunctions2d/distfunctions2d.htm
3D SDF functions	https://iquilezles.org/www/articles/distfunctions/distfunctions.htm
Intersection functions	https://iquilezles.org/www/articles/intersectors/intersectors.htm
Raymarching SDFs	https://iquilezles.org/www/articles/raymarchingdf/raymarchingdf.htm
SDF bounding volumes	https://iquilezles.org/www/articles/sdfbounding/sdfbounding.htm
Numerical normals for SDFs	https://iquilezles.org/www/articles/normalsSDF/normalsSDF.htm
fBM	https://iquilezles.org/www/articles/fbm/fbm.htm
Gradient noise derivatives	https://iquilezles.org/www/articles/gradientnoise/gradientnoise.htm
Voronoise	https://iquilezles.org/www/articles/voronoise/voronoise.htm
GPU conditionals	https://iquilezles.org/articles/gpuconditionals/
XOR of two SDFs	https://iquilezles.org/articles/sdfxor/
Domain repetition	https://iquilezles.org/articles/sdfrepetition/

Retained UGTS technical basis

• UGTS-KC 3.9 KC Elizabeth Vector Game Runtime: retained vector/game/project/browser architecture and deterministic fixed-step world.

• UGTS-KC Two Hands 3.0: retained scene, geometry compiler, spatial query, hand interaction, event proposal/commit and replay shell.

• UGTS-KC K-Kij-T Grove 3.9.2 complete package: immediate implementation baseline for the new subversion.

THIRD-PARTY NOTICE

Inigo Quilez article titles and public URLs are referenced for technical survey and attribution. This implementation is an original bounded re-expression in the UGTS architecture. It does not claim endorsement, ownership of the source articles or relicensing of their content.

Final formal definition

UGTS-KC 3.9.3 IQ Field is a finite, query-first field calculus in which exact distance primitives, conservative estimates, CSG, domain repetition, analytic intersections, bounded sphere tracing, deterministic procedural signals, footprint filtering, texture projection and color transforms are typed capabilities upstream of the retained UGTS event authority. Every iterative path has a budget and status. Every repeated or procedural state retains address and seed lineage. Presentation never silently becomes authoritative state.